Loudspeaker project, System D: work plan

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Based on `34870_Project_Guide_2026.pdf`. Tick the boxes as we go, and put a name on each step.

How to use this

The project follows the guide's five tasks. Each step below says what goes in (which file we already have), what we do with it, what comes out, and how we check it before moving on. Do the steps in order inside a task; Tasks 1 and 2 can run in parallel. Every result that ends up in the quiz or the report gets a figure in `Project/figures/` and a line in `Project/report/Project_report.tex`.

Date	Milestone
Sun 8 Nov	Project quiz part 1: LTspice models and the Lab D/E data in them (Tasks 1 and 2)
November	IEC listening room booked for all of November (Task 4)
Mon 30 Nov	Project quiz part 2: final results and discussion
Thu 3 Dec	Demonstration in room 019: bring the passive board and the DSP coefficients

What we already have

Data	File (labs repo)
Woofer impedance, free air and closed box	`LabD/data/labD_woofer_free.mat`, `labD_woofer_closed.mat`
Woofer impedance, one vent at 160/200/240 mm	`LabD/data/labD_vent_L160/L200/L240.mat`
Near field of cone and vent, each length	`LabD/data/labD_nf_cone_L*.mat`, `labD_nf_vent_L*.mat`
Midrange and tweeter impedance Z_i	`LabD/data/labD_midrange.mat`, `labD_tweeter.mat`
Anechoic p/V of all three units, 0° and 30°	`LabE/data/labE_system_0.mat`, `labE_system_30.mat`
Woofer data sheet	`LabD/datasheets/`
Course tools	`Project/matlab/course/` (`Matlab2LTspice.m`, `DigitalCrossover.m`, `ventbox`)
Course LTspice examples	`Project/ltspice/course/passive/`, `digital/`
One loader for all of it	`Project/matlab/load_systemD.m`
Ready to use (made from the data above)	
LTspice data files, names already unique	`Project/data/Zi_mid.txt`, `Zi_tw.txt`, `TF_\{woof,mid,tw\}.txt`, `TF30_*.txt` (made by `matlab/export_systemD.m`, $d = -1.15\,\mathrm{m}$)
The same data as CSV (Python, Excel)	`Project/data/csv/`
One test schematic per unit	`Project/ltspice/units/Unit_woof.asc`, `Unit_mid.asc`, `Unit_tw.asc`
Empty system skeleton (Figure 1)	`Project/ltspice/System_template.asc`

The unit and system files are written by `Project/ltspice/gen_templates.py`; `python3gen_templates.py--verify` runs LTspice and confirms each unit reproduces the measured Z_i and p/V (to 0.04 dB). Open them in LTspice straight from the repo: the `.inc` paths are relative.

Known problems with the data (see `Project/README.md`): Z already uses $R = 32.9\,\Omega$ and $V_{\mathrm{amp}} = 2 \cdot \mathrm{AI0}$; temperature was not recorded (assume 20 °C); the UMIK channel is unusable above 18.4 kHz; `labE_system_30_bad.mat` is rejected; the woofer TF in Lab E was taken with the vent plugged.

Task 0: Before we start

☐ Agree who owns which task, and book lab time with Group 4 (we share System D). [who: _____]

☐ Measure what we never wrote down, next time we are at the box: inner diameter of *both* vents, how the telescopic length is defined (and its min/max), whether there was filling in the box during the Lab D runs, and the room temperature. [who: _____]

Check: Task 1 needs the second vent's diameter to model "two vents". Without it we can only model the one we measured.

☐ Everyone runs `S=load_systemD` in MATLAB once, so we know the data loads on every PC. [who: _____]

Task 1: Low-frequency tuning of the woofer box

Lectures 7 and 8, Lab D. Goal: choose closed or vented, the vent mass (none, one, two vents) and filling, from an LTspice analogy that has been validated against our near-field measurements.

1.1 Thiele–Small parameters

☐ Run `LabD/matlab/analyse_labD.m` and copy f_S, Q_{MS}, Q_{ES}, Q_{TS}, V_{AS}, R_E into a table next to the data-sheet values. [who: _____]

Output: T-S table for the report (section 2.1).

Check: f_c/f_S should equal Q_{TC}/Q_{TS} (the script prints both). Explain why our V_{AS} differs from the data sheet.

☐ Derive the circuit elements M_{MS}, C_{MS}, R_{MS}, Bl from them (S_D from the data sheet), and decide how much of M_{MS} is air load (Lecture 7: $M_{MD} = M_{MS}-$ air mass). [who: _____]

1.2 LTspice analogy circuit

☐ Draw the woofer-in-box circuit in LTspice (Lecture 7 slide 27 / Lecture 8): electrical, mechanical and acoustical domain, front radiation impedance, box compliance C_{AB}. Start from the closed box. [who: _____]

Check: Simulated $|Z_E|$ against `labD_woofer_closed.mat`: same peak frequency and height.

☐ Add the vent: acoustic mass $M_{AP} = \rho(L + \Delta L)/S_P$ in parallel with C_{AB}, plus a leak/loss resistance. Write down which end correction we use and why. [who: _____]

Check: Simulated $|Z_E|$ against `labD_vent_L160/200/240.mat`: both peaks and the minimum between them ($\approx f_B$). If the formula does not move f_B the way the measurement does, write that down; it tells us something about the vent geometry.

☐ Add outputs: far-field pressure $\propto j\omega(U_D - U_P)$ and the near-field pressures of cone and vent ($\propto j\omega U/a$). [who: _____]

1.3 Validation against the near field

☐ Plot simulated against measured near field (cone and vent) for each vent length, using `labD_nf_*`. [who: _____]

Output: One figure per length, or one figure with three panels.

Check: Cone notch at f_B, vent peak around f_B, crossing point of the two curves. Explain any offset (mic position, room modes below 120 Hz, the vent mic picking up the cone above ≈60 Hz).

1.4 Choose the tuning

☐ With the validated circuit, step through the options the guide allows: closed; one vent at different lengths; two vents in parallel ($M_{AP,1} \parallel M_{AP,2}$); filling (larger effective volume, more damping). [who: _____]

☐ For each one, read off f_3 and the ripple of the low-frequency response. [who: _____]

Output: A table option → f_B, f_3, ripple, and one overlay plot.

☐ Pick one and argue for it (bandwidth against ripple, cone excursion near f_B, what the standard alignments in Lecture 8 would ask for, and why we cannot reach them). [who: _____]

Task 2: Modelling the crossover filters

Lecture 9, Lab D and Lab E.

2.1 Measured data into LTspice

☐ Read `matlab/export_systemD.m` and understand what it does: it calls `Matlab2LTspice.m` for Z_i of midrange and tweeter (no distance) and the TF of all three units (same d for all), then renames the first line of each file (`Vspkr_mid`, `Pout_mid`, ...) because duplicate names break the simulation. Try another d if the phase is hard to read, and re-run it. [who: _____]

☐ Open `ltspice/units/Unit_mid.asc`, `Unit_tw.asc`, `Unit_woof.asc` and run them. [who: _____]

Check: LTspice $|Z|$ and phase equal Lab D; `V(Pout_*)` equals Lab E's p/V (the script checks this, but look at the plots yourselves and put one in the report).

☐ Note: the course's `...andexamplefilter.asc` is drawn as a **high-pass** (C in series, L to ground) although the guide's Figure 2 calls it a low-pass. [who: _____]

2.2 Choose the two crossover frequencies

☐ Overlay the three measured responses (Lab E, 0° and 30°) and the impedances. [who: _____]

☐ Woofer–midrange: where both are still smooth, and where the woofer has not started to beam (compare 0° and 30°). [who: _____]

☐ Midrange–tweeter: well above the tweeter resonance (Lab D), and where the midrange still behaves off axis. Note the level difference between the two units. [who: _____]

Output: Two frequencies with one line of justification each.

2.3 Digital crossover (woofer / mid-high)

☐ Run `DigitalCrossover.m` for the LPF and the HPF at our frequency; decide the order. [who: _____]

Check: The function makes Butterworth sections ($n = 4$ gives $Q = 0.541$ and 1.307). If we want Linkwitz–Riley, cascade two $n = 2$ sections at the same f_0, and say so in the report.

☐ Paste the Laplace strings into the E sources of `ltspice/course/digital/`, woofer branch first, and check the filter on its own. [who: _____]

2.4 Passive crossover (midrange / tweeter)

☐ Choose topology and order; compute the values for the nominal impedance. [who: _____]

☐ Round to the standard values in Appendix 1 (no combining components). [who: _____]

☐ Simulate each filter loaded with a resistor, then with the real Z_i, and compare. [who: _____]

Check: Does the driver impedance shift the corner? Add a Zobel or a series resistor if it does.

☐ Match the tweeter level to the midrange (L-pad or series resistor). [who: _____]

☐ Tolerance check: step the component values by their tolerance and look at the sum. [who: _____]

2.5 Complete system

☐ Start from `ltspice/System_template.asc`: it is Figure 1 already wired, with gain-1 biquad slots and plain wires where the passive filters go. Then wire all three units as in the guide's Figure 1 (digital LPF → woofer; digital HPF → passive LPF/HPF → mid/tweeter), and sum the three outputs with series controlled sources. Try polarity ±1 per unit. [who: _____]

☐ Add the woofer/mid level difference as a gain (this becomes the DSP output gain). [who: _____]

☐ Iterate (frequencies, orders, polarity, levels) until the on-axis sum is flat and the 30° sum is reasonable. Keep a log of each iteration and why it changed. [who: _____]

Output: Final LTspice schematic, on-axis and 30° sum, iteration log.

Quiz part 1 (Sun 8 Nov)

☐ Everyone answers individually, but compare figures beforehand: T-S table, validated box circuit, near-field comparison, LTspice Z_i/TF checks, first full-system sum. [who: _____]

Task 3: Implementing the crossovers

☐ MiniDSP: paste the `biquad_DSP` strings into Xover of outputs 1 and 2 (everything else bypassed); set the output gains from 2.5. [who: _____]

☐ Passive board: mount our standard values; photograph it; note which parts we used. [who: _____]

☐ Anechoic measurement of the complete system (vents may be covered), on and off axis. [who: _____]

Check: Measured against simulated. Explain the differences and decide if another iteration of Task 2 is needed.

Task 4: Listening test (IEC room, November)

☐ Read Lecture 10 (and Bech & Zacharov if needed) and write a short procedure: what we compare, in what order, how long, listening position, level. [who: _____]

☐ Choose the programme material (a few tracks with clear bass, voice, cymbals). [who: _____]

☐ Write the questionnaire (fixed attributes and scale), and find subjects. [who: _____]

☐ Run the test, collect the answers, summarise them, and decide what to change. [who: _____]

Task 5: Quiz part 2 and demonstration

☐ Final results and discussion in quiz part 2 (Mon 30 Nov), individually. [who: _____]

☐ Demonstration Thu 3 Dec: passive board and DSP coefficients ready, plus the plots we want to show. [who: _____]

☐ Fill in `Project_report.tex` section by section as each task finishes. [who: _____]